

Client Change Request (CCR)**CCR Number:** CCR-009**Client:** Atlantic Shipping & Logistics**Submitted By:** Karen Brooks, Shipping Operations Manager**Priority:** Medium**Requested Delivery:** End of Today's Sprint**Assigned Developer:** _____**Date Assigned:** _____

Business Background

Atlantic Shipping & Logistics prepares thousands of packages for shipment every week. Before packages are loaded onto trucks, employees measure each shipping box to ensure that it meets carrier requirements and to determine the amount of packaging material needed.

Currently, employees use handheld calculators to determine the volume and surface area of each box. This process is repetitive and occasionally results in incorrect calculations, leading to shipping delays and wasted packaging materials.

The Shipping Operations Manager would like a computer program that accepts the dimensions of a shipping box and calculates both its volume and surface area. These calculations should be performed independently so that the results can be reused by other parts of the shipping software in future versions of the system.

At this time, the program only needs to process one shipping box during each execution.

Customer Requirements

The employee should be able to enter:

- Length of the shipping box
- Width of the shipping box
- Height of the shipping box

The program should calculate:

- The volume of the box
- The surface area of the box

After the calculations have been completed, the program should display:

- The dimensions entered by the employee
- The calculated volume
- The calculated surface area

The report should be neatly formatted and easy for employees to verify.

Acceptance Criteria

The customer will consider the software complete when all of the following are true.

- ☐ Employee can enter the box length.
 - ☐ Employee can enter the box width.
 - ☐ Employee can enter the box height.
 - ☐ Program correctly calculates the volume.
 - ☐ Program correctly calculates the surface area.
 - ☐ Program displays all entered dimensions.
 - ☐ Program displays the calculated volume.
 - ☐ Program displays the calculated surface area.
 - ☐ Program produces a clear, professional report.
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Out of Scope

The customer is **not** requesting:

- Processing multiple boxes
- Saving measurements to a file
- Calculating shipping costs
- Determining package weight
- Determining dimensional weight
- Selecting shipping carriers
- Printing shipping labels
- Tracking packages
- Graphical user interfaces (GUI)

If these features appear in the software, they will not be evaluated.

Example Scenario 1

Employee Input

Length: 10

Width: 8

Height: 6

Expected Output

Atlantic Shipping & Logistics

Package Measurement Report

Length : 10.00

Width : 8.00

Height : 6.00

Volume : 480.00 cubic units

Surface Area : 376.00 square units

Example Scenario 2

Employee Input

Length: 5

Width: 5

Height: 5

Expected Output

Atlantic Shipping & Logistics
Package Measurement Report

Length : 5.00

Width : 5.00

Height : 5.00

Volume : 125.00 cubic units

Surface Area : 150.00 square units

Example Scenario 3

Employee Input

Length: 20.5

Width: 15.0

Height: 12.5

Expected Output

Atlantic Shipping & Logistics
Package Measurement Report

Length : 20.50

Width : 15.00

Height : 12.50

Volume : 3843.75 cubic units

Surface Area : 1630.00 square units

Notes from the Customer

"I don't care how the calculations are performed internally. I simply want the employee to enter the dimensions of a box and receive accurate measurements for both its volume and surface area. As our software grows, I'd like these calculations to be reusable by other parts of the system."

Developer Notes

This section intentionally left blank.

Students will complete the software requirements during class before beginning implementation.

Questions for the Customer

Before writing any code, list any questions you would ask the customer to clarify the request.

Examples:

- 1.

Reminder

The customer has intentionally described the problem using **business language**, not programming terminology. During class, your first task is to translate these customer requirements into **software requirements** before writing any C code.

For this assignment, you should use only the C language features that have been covered through **Lesson 8**, including:

- Variables and constants
- Input/output
- Formatted output
- Selection statements (if and switch)
- Loops
- Functions
- Local and global variables
- Pointers
- Passing pointers as function arguments
- Returning values from functions

Do **not** use C language features that have not yet been introduced in the course.